

**Stochastic Programming.**  
**Problems session 3**

11. John Thedoor, president of a prestigious football team from Barcelona wants to change some of the players. He considers to sell up to  $m$  players, each of them reporting  $r_j$  €,  $j = 1, \dots, m$ , respectively. He also considers to buy up to  $n$  new players, with a cost of  $s_i$  €,  $i = 1, \dots, n$ , respectively. The total money spent on new transfers may not exceed the total money obtained by leaving players.

The next season each of the new players will have a goal and defense capabilities  $g_i$  and  $d_i$ ,  $i = 1, \dots, n$ , respectively.  $g_i$  and  $d_i$  are random variables. It has been estimated that for the team to be the champion next season, the global goal and defense capabilities provided by new players (computed as the sum of individual capabilities) should be greater than or equal to  $\xi_g$  and  $\xi_d$ , both random variables. It is estimated that, for every unit that the global goal capability provided by new players is below  $\xi_g$  the chances of being the champion are reduced by 1 point. Similarly, for every unit the global defense capabilities provided by new players is below  $\xi_d$ , the chances of being champion are reduced by 1.5 points.

- (a) Formulate the stochastic problem to be solved by the president of the club, to decide which players to sell and which to buy, such that the chances of not being the champion are minimized.  
**Tip:** Use binary variables to know which players to sell or buy.
- (b) Which would be the number of scenarios of the extensive of the above problem, considering independent random variables, that  $g_i$  and  $d_i$   $i = 1, \dots, n$  can take 4 different values, and that each of the variables  $\xi_g$ ,  $\xi_d$  may have 5 different values?
12. To overcome the economic crisis a country decides to establish two types of grants to encourage consumption: direct (giving nonrepayable money) and indirect (zero interest loans to return in a few years) subsidies. The total budget for the two types of loans can not exceed 5000 million €. The costs for the country are the total amount in direct grants, and a 10% of indirect ones (by loss of interest). It is estimated that each € in direct grants will mean  $\xi_1$  € to the country economy; similarly, each € in indirect grants will mean  $\xi_2$  € to the country economy. Let us assume that  $\xi_1$ ,  $\xi_2$  are random variables of expected value greater than 1. If this policy does not provide more than 6000 millions € of benefit to the country economy, we will assume that the country incurs in a large and fixed cost of  $F$  millions € (in addition to the costs associated to the grants). On the other hand, if the benefits are greater than 7500 million €, we will assume that there are no additional costs to the country. Finally, if the benefits are between 6000 and 7500 million €, we will assume a linear cost of  $v$  million € for each million € below 7500 million €. Assume the fixed cost is higher than the maximum linear cost, i.e.,  $F > 1500v$ .
- (a) Formulate the stochastic optimization problem to be solved by this country (of course, we are assuming—and this is science fiction—it is a serious country with a honest and well prepared government).
- (b) Write the extensive form, assuming that both random variables are totally dependent with values  $\xi_{1_i}, \xi_{2_i}, i = 1, \dots, s$  and probabilities  $p_i, i = 1, \dots, s$ . How many scenarios has the resulting problem?
13. The multinational computing corporation HAL has to sign an agreement to deliver  $x_1$  and  $x_2$  units of computer models 1 and 2 to its customers for the next three months. Each unit means some hours for manufacturing and testing, according to this table:

|         | manufacturing | testing   |
|---------|---------------|-----------|
| Model 1 | 5 hours       | 1 hour    |
| Model 2 | 4 hours       | 1.5 hours |

The total number of available hours at the company for the next three months, for manufacturing and testing, are independent random variables  $\xi_1$  and  $\xi_2$ , respectively. Each of the agreed computers

means a benefit of  $b_1$  and  $b_2 \in$ , depending of the model; this amount will be paid by the customers at the time of the agreement, before the manufacturing has started. After the three months, for each non provided computer the company will pay a penalty of  $c_1$  and  $c_2$  to the customers.

Formulate a stochastic optimization problem to obtain the optimal number of computers per model to be set in the agreement.

14. (a) The company “ApplePie” wants to launch a new computer gadget, and it must decide the production capacity of this product ( $p$  units) and the selling price ( $x \in$ /unit). The selling price will be between minimum and maximum prices  $l$  and  $u$ . The production capacity has a cost  $c \in$ /production unit (due to the size of the production plants). Future sales depend on the demand, which is a random variable  $\xi$ ; they also depend on the production capacity (it can not sell more than produced). The company wants to avoid risk: it has been estimated that the distribution of  $\xi$  is a function of the selling price  $x$ , so that the higher the price, the lower the demand, and vice-versa; for this reason the company considers that if in the future the demand is less or equal than  $t$  customers, a fixed risk of  $r \in$  has to be considered in the second stage (risk associated to the lack of interest in this gadget). Formulate the stochastic optimization problem to be solved by the company for the optimal  $p$  and  $x$ , to minimize the capacity production costs, maximize the future benefits, and minimize the risk.
- (b) Formulate the following variant with probabilistic constraints to the above problem . The objective function is  $\min \beta_1 p - \beta_2 x$ , where  $\beta_1$  and  $\beta_2$  are production cost and selling benefit, respectively. The selling price is bounded by minimum and maximum prices  $l$  and  $u$ . We assume demand  $\xi$  has a continuous uniform distribution between  $m/x$  and  $3m/x$ ,  $m$  being a predefined parameter. Note that the distribution of  $\xi$  depends on the price (the higher the price, the lower the demand, and vice-versa). We will impose two probabilistic constraints: (1) the probability of demand being less or equal to  $t$  customers has to be less or equal to  $\alpha_1$ ; (2) the probability of capacity production  $p$  being greater than or equal to demand has to be greater than or equal to  $1 - \alpha_2$  ( $t$ ,  $\alpha_1$  and  $\alpha_2$  are problem parameters). Formulate the problem and simplify the probabilistic constraints until obtaining an optimization problem. Assume  $t$  is between  $m/x$  and  $3m/x$ , the demand bounds.